

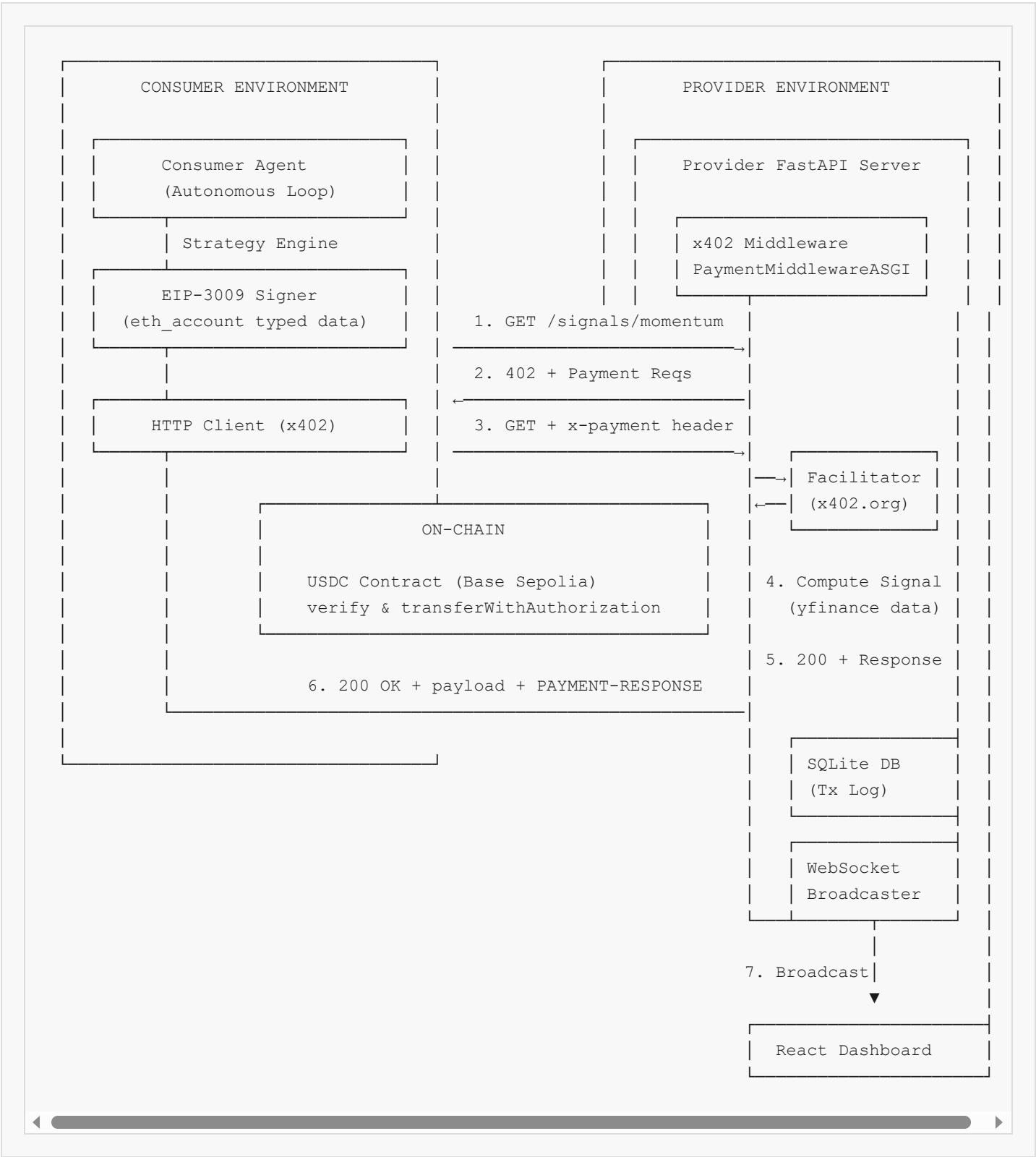
1. SYSTEM OVERVIEW

QuantMesh is a high-frequency, pay-per-query financial data marketplace designed for autonomous AI agents. It completely decouples financial data consumption from traditional subscription models, utilizing the x402 payment standard to facilitate real-time, API-level micropayments. By leveraging EIP-3009 (

```
transferWithAuthorization
```

) and Arc's sub-cent network economics via Base Sepolia, QuantMesh solves the fundamental infrastructure problem of M2M (Machine-to-Machine) commerce: the incompatibility of L1 gas costs with sub-cent commercial transactions. QuantMesh makes it economically viable to price distinct market signals at highly granular levels (e.g., \$0.001 to \$0.005), realizing net margins of ~99.5%.

2. ARCHITECTURE DIAGRAM (ASCII)



3. PAYMENT FLOW

The following describes the exact transaction settlement protocol sequence utilizing EIP-3009:

1. **Initial Request:** The Consumer Agent issues an HTTP GET to a protected endpoint (e.g.,

```
/signals/momentum/BTC-USD
```

).

2. **Rejection & Challenge:** The Provider's

```
PaymentMiddlewareASGI
```

detects the missing

```
PAYMENT-SIGNATURE
```

header. It aborts the request, returning an

```
HTTP 402 Payment Required
```

payload exposing the demanded price and recipient (

```
payTo
```

address).

3. **EIP-3009 Framing:** The Consumer Agent derives the required payment option. It constructs an EIP-712 domain separator bound to the USDC token contract on Base Sepolia (

```
eip155:84532
```

).

4. **Offline Signing:** The Consumer explicitly constructs the

```
TransferWithAuthorization
```

typed data structure (including a highly entropic 32-byte nonce, the agreed amount, and

```
validBefore
```

expiry window setup 300 seconds forward) and signs it offline using

```
eth_account.sign_typed_data
```

.

5. **Request Retry:** The Consumer retries the original HTTP GET request, now attaching the constructed x402 payload bearing the EIP-3009 signature within the

```
PAYMENT-SIGNATURE
```

header.

6. **Provider Verification:** The Provider

```
PaymentMiddlewareASGI
```

traps the request, proxying the signature to the facilitator (

```
x402.org
```

).

7. **Settlement:** The facilitator wraps the signature, calls

```
transferWithAuthorization
```

on the designated USDC contract on-chain, and awaits absolute finality.

8. **Resolution:** Upon facilitator confirmation, the middleware resolves the request path, invokes the underlying FastAPI application route, and bundles the retrieved on-chain

```
transactionHash
```

within the

```
PAYMENT-RESPONSE
```

header sent alongside the

```
HTTP 200 OK
```

response payload.

4. SIGNAL COMPUTATION LAYER

Signals are deterministically evaluated. To mitigate excessive third-party API rate limits, quantitative computations are subject to a high-performance in-memory cache utilizing

```
cachetools
```

configured with a 30-second TTL.

- **Momentum**

- **Formula:** $ROC = \frac{P_t - P_{t-14}}{P_{t-14}}$
- **Logic:** Calculates the 14-day Rate of Change evaluating near-term bullish/bearish continuation velocity.

- **Volatility**

- **Formula:** $\sigma_{ann} = \text{std}(\ln(\frac{P_t}{P_{t-1}})), 20) \times \sqrt{252}$
- **Logic:** Computes the 20-day annualized realized historical volatility through logarithmic daily returns over a standard 252-day continuous trading period.

- **Sentiment**

- **Formula:** $\frac{N_{positive} - N_{negative}}{N_{total}}$ bounded to $[0, 1]$
- **Logic:** Applies weighted NLP deterministic modeling over trailing

```
yfinance
```

headline metadata mapping institutional sentiment.

- **Arbitrage Spread**

- **Formula:** $Z_{spread} = \frac{P_1 - P_2}{(P_1 + P_2) / 2}$
- **Logic:** Evaluates the normalized differential cross-pair statistical variance, exposing convergence divergence.

5. AGENT STRATEGY LOGIC

The consumer agent deploys an autonomous stochastic loop operating cyclically every 3 seconds.

1. **Signal Permutation:** For each cycle, the

```
SimpleStrategy
```

pseudorandomly fetches between 1 and 4 distinct signals ranging across indices (

```
BTC-USD
```

```
,
```

```
ETH-USD
```

```
,
```

```
SOL-USD
```

```
,
```

```
AAPL
```

```
,
```

```
MSFT
```

```
,
```

```
SPY
```

).

2. **Signal Synthesis:** A composite probabilistic score calculates confidence based on aggregated weighted inferences:
 - Momentum > +3% assigns Bullish skew.
 - Low volatility assigns lower risk premiums, improving confidence parameters.
 - High institutional sentiment dictates macro-level trajectory direction.

3. **Execution Condition:**

- IF

```
composite_score > 0.65
```

: Emit

```
BUY
```

execution protocol.

- IF

```
composite_score < -0.65
```

: Emit

```
SELL
```

execution protocol.

- ELSE: Abort execution (

```
HOLD
```

).

4. **State Subsystem:** Maintains volatile in-memory ledger states dynamically estimating mock P&L accumulation parameters.

6. ECONOMIC MODEL

Sub-cent transaction architecture depends rigidly on ultra-low L2 block baselines. The QuantMesh sub-cent profitability thesis is fundamentally impossible strictly isolated on Ethereum Mainnet.

Metric	Arc/Base Sepolia	Ethereum Mainnet
Gas per tx	~\$0.000000336	~\$1.50
Revenue per tx	\$0.001-\$0.005	\$0.001-\$0.005
Net margin	~99.5%	-69,130%
172 tx gas cost	\$0.000058	\$258.00
172 tx revenue	~\$0.35	~\$0.35

Mathematical Proof of Impossibility (Ethereum L1):

Let S equal minimum deterministic signal pricing ($\$0.001$).

Let G equal normalized deterministic gas limit settlement parameters ($\$1.50$).

$\text{Margin} = \frac{S - G}{S} \times 100 = \frac{0.001 - 1.50}{0.001} \times 100 = -149,900\%$.

Gas expense strictly outpaces absolute revenue by orders of magnitude.

7. DATABASE SCHEMA

The SQLite persistence layer establishes a rigorous ledger format defining exact on-chain mapping to agent behaviors.

```
CREATE TABLE IF NOT EXISTS transactions (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    tx_hash TEXT UNIQUE NOT NULL,  
    payer TEXT NOT NULL,  
    endpoint TEXT NOT NULL,  
    amount REAL NOT NULL,  
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

8. SECURITY CONSIDERATIONS

Due to its decentralized and distributed M2M nature, QuantMesh implements stringent payment security protocols:

- **Nonce Entropic Density:** Mandates highly unpredictable cryptographic values initialized strictly via

```
os.urandom(32)
```

unique per every transaction iteration.

- **Bounded Vulnerability Window:** Leverages a robust

```
validBefore
```

authorization sequence guaranteeing unexecuted payments expire completely post-300 seconds resolving indefinite exposure.

- **Deterministic Isolation:** Consumer + Provider isolated Private Keys securely enveloped via restricted unversioned

```
.env
```

profiles.

- **x402 Replay Nullification:** Enforces strictly absolute semantic deduplication across the x402 facilitator relay verifying isolated nonce-specific utilization.